

Principal component analysis for network RM

Fedor Nikitin

June 29, 2013

1 Introduction

Network revenue management exhibits complex dynamics.

Detecting changes in airline network is important...

2 Network optimization

Network revenue management problem could be formulated as dynamic programming problem. However due to curse of dimensionality it is intractable. Different approximations has been proposed [?].

Simplest and widely used in practice approximation is Determenistic Linear Program (DLP) which is

$$R = p^T x \rightarrow \max, \quad (1)$$

$$Ax \leq c, \quad (2)$$

$$0 \leq x \leq \mu, \quad (3)$$

where $p^T \in \mathbb{R}^n$ is price vector, $A \in \mathbb{R}^{m \times n}$ is incidence matrix, $c \in \mathbb{R}^m$ is capacity vector and $\mu \in \mathbb{R}^n$ is expected demand.

Airline demand experiences seasonal fluctuation and it is important to adapt for seasonal fluctuations. In practice the whole planning period is split into the parts which are called planning windows.

3 Feature selection

We are given OD demand vector $\mu(t)$ defined on time interval $[t_0, T]$. Our goal is to define points in time when demand vector $\mu(t)$ experiences significant changes.

3.1 Passenger flow

Demand flown out of vertex i is given by

$$\mu_i^{out}(t) = \sum_{j \in V, k \in F} \mu_{ijk}(t).$$

Demand flown in vertex i is

$$\mu_j^{in}(t) = \sum_{i \in V, k \in F} \mu_{ijk}(t).$$

3.2 Revenue flow features

$$R_i^{out}(t) = \sum_{j \in V, k \in F} p_{ijk} \mu_{ijk}(t)$$

$$R_i^{in}(t) = \sum_{i \in V, k \in F} p_{ijk} \mu_{ijk}(t)$$

3.3 Travel pattern feature

$$S(t) = \{s_{ij}(t)\} \in \mathbb{R}^{n \times n}$$

$$s_{ij}(t) = \frac{\sum_{k \in F} \mu_{ijk}(t)}{\sum_{j \in V, k \in F} \mu_{ijk}(t)}$$

4 Principal component analysis

5 Segmentation

6 Experiments